Optimizing Congestion in High-Load Singapore MRT Stations Using Mathematical Passenger-Flow and Queuing Models

Introduction and Background

Singapore's Mass Rapid Transit (MRT) system is a critical component of the nation's urban mobility network, supporting millions of passenger journeys daily. As ridership continues to increase, congestion has become a persistent operational challenge, particularly at stations with high passenger throughput and complex transfer functions.

Recent data-driven analysis based on station-level origin–destination passenger-load data demonstrates that demand is highly uneven across the MRT network, with a small number of stations consistently carrying disproportionately large passenger volumes during peak periods (Tan, 2025). The analysis shows that stations such as Jurong East MRT Station, Raffles Place MRT Station, and Tanjong Pagar MRT Station rank among the busiest stations during both morning and evening peaks, indicating persistent congestion pressure rather than isolated crowding events.

These empirical passenger-load patterns provide quantitative evidence that congestion in the MRT system is structural and recurrent, particularly at interchange stations and central business district stations with strong directional commuter flows (Tan, 2025).

Literature Review

Congestion is an existing problem in Singapore's MRT stations. Online searches using keywords such as "Singapore," "MRT," "congestion," and "crowd" yield a substantial number of results over a long-time span and at high frequency; typical examples include reports by Yee (2025), Romero (2025), and Chelvan (2026).

However, the severity of congestion is not uniform across stations. By analyzing data from the Land Transport Authority DataMall and other sources, Tan (2026) found that passenger flow distribution is highly uneven, with a small number of MRT stations handling a disproportionately large share of passengers. A consistent result is reported by Admin (2025), which indicates that 20% of the daily MRT passenger flow is carried by the 10 busiest stations. Therefore, station-level optimization targeting the most congested stations is potentially beneficial. GIven the above conclusion, this study has chosen Raffles Place MRT station as it is one of the most congested MRT stations in Singapore.

Government efforts have been made to alleviate peak-period pressure in MRT stations. In one case, the Land Transport Authority increased train frequency to accommodate high passenger volumes (Tan, 2020). The impact of another potential solution—providing alternatives such as buses and taxis—is relatively limited, as the capacity of the MRT system far exceeds that of road transport (Tang, 2017). For instance, a single MRT train has a capacity equivalent to approximately 15 buses. Station-level flow management measures are also used; however, they are currently part of incident management plans and are typically implemented only during special events or large crowd gatherings (Ministry of Transport, 2022). To date, no research or public documentation appears to have proposed station-level crowd management as a routine optimization strategy for congestion.

A wide range of studies with similar objectives and characteristics have been conducted on stations worldwide, providing valuable theoretical references. Based on previous research, three major approaches to describing passenger flow can be identified: mathematical modeling, analogue simulation, and a combination of both. Xu et al. (2014) developed a mathematical model to characterize station capacity and passenger volume. Using simulation techniques, Lee and Marinov (2022) generated several practical recommendations for congestion mitigation. Similar results were achieved by King et al. (2026), who optimized a Canadian subway station using simulation software. Zhang et al. (2024) combined mathematical modeling and simulation, employing analytical models to diagnose existing issues and simulation tools to evaluate optimization strategies. Overall, existing studies demonstrate that integrating analytical and simulation-based approaches is an effective methodology for understanding and mitigating station-level congestion.

Research Aims, Objectives, and Questions

3.1 Research Aim

To develop and evaluate mathematically grounded operational strategies that reduce congestion and improve passenger-flow efficiency at Singapore's busiest MRT stations.

3.2 Research Objectives

1. To justify the selection of high-congestion MRT stations using empirical passenger-load data (Tan, 2025);
2. To model congestion mechanisms using established queuing and crowd-flow formulations (Xu et al., 2014; Albi et al., 2020);
3. To design operational congestion-mitigation strategies grounded in mathematical models;
4. To quantitatively evaluate the effectiveness of these strategies using established performance indicators (Zhang et al., 2024).

3.3 Research Questions

1. Does congestion occur at MRT stations during peak hours, and how is it specifically manifested?
2. Can mathematical passenger-flow models inform effective operational congestion-mitigation strategies?
3. How do optimized operations compare with existing configurations in terms of congestion indicators?

Research Design and Methodology

4.1 Field Visit

To complement the theoretical modelling framework and verify the practical relevance of the research problem, a field visit will be conducted at selected MRT stations during peak commuting periods. The purpose of the field visit is to observe real passenger-flow patterns, identify potential congestion points, and compare these observations with the congestion mechanisms suggested by previous research.

Observations could be carried out at Raffles Place MRT station from 5:00pm to 7:00pm. The observation includes three parts:

1. The average time spent queuing in front of the gates in both normal time and specific peak hours

2. The average time spent queuing in front of one specific escalator in both normal time and peak hours.

3. The average time spent transformation from North South Line to East West Line.

The difference could be computed and compared to prove the severity of congestion.

This station, as observation subject is selected because passenger-load data indicates that it consistently experience some of the highest ridership levels in the MRT network (Tan, 2025). Besides, with its central location next to CBD area and essential function as the joint point of East West Line and North South Line, Raffles Place stands out as a ideal target for congestion optimization.

4.2 Data Sources

1. Station-level passenger-load data (AM and PM peaks) which could be retrieved from official API offered by LTM Datamall or published materials;
2. Station layout characteristics (platforms, corridors, escalators);
3. Model parameters adopted directly from cited literature.

4.3 Modelling Framework

Bottlenecks are modeled using queuing theory, with expected queue length and waiting time calculated following Xu et al. (2014). For the real situation in an MRT station, the Bottleneck-points could be the ticket check entrance and crowded escalators or elevators during peak hours. Overall pedestrian movement is analyzed using crowd dynamics conservation principles (Albi et al., 2020). Congestion severity in testing model could be quantified using performance indicators such as the time consumption of total travel time estimated as:

$$t_i = \frac{d_i^{ep}}{v_i} + \frac{h_i^{ep}}{v_i \cdot \sin\alpha} + \sum_{j=1}^{k} t_{i,j}^{q} + \sum_{j=1}^{k} t_{i,j}^{s}$$

where t_i is the total journey time for a passenger marked as i, $t_{i,j}^{q}$ and $t_{i,j}^{s}$ show the queuing time and service time at service facility j respectively. The first two terms in the right side of the equation are given as the walking and vertical travel times (Zhang et al., 2024).

In addition, the congestion severity in other crucial part of the station could be measured using the crowding model as applied by Zhang et al. (2024) based on real operational variables:

$$d_{i+1} = \frac{q_i + I_{i,i+1}(\alpha_{in} + \alpha_{tr}) + 2\beta_{al}}{w_p \cdot l_p - n_e w_e l_e - n_c - 4n_s} \cdot \gamma$$

where $\boldsymbol{d_{i+1}}$ measures the maximum platform density ($\boldsymbol{pax/m^2}$), $\boldsymbol{q_i}$ stands for the number detained passengers, $\boldsymbol{I_{i,i+1}}$ represents the train headway in minute, $\boldsymbol{\alpha_{in}}$ and $\boldsymbol{\alpha_{tr}}$ show the entering and transfer-in rates, and at end $\boldsymbol{\gamma}$ represents the spatial density disequilibrium factor. Besides all these participants, other variables in the equation are actually physical layout adjustments accessible from published constructional maps about our studying stations.

The space model of an MRT station could be simplified into 2 dimensional grid map including multiple layers of a station.To assess the effectiveness of congestion-management measures, Zhang (2024) compares these indicators across different scenarios, such as baseline operations versus adjusted flow-control strategies. Improvements are demonstrated when optimized scenarios produce lower peak passenger density, shorter queues, and reduced total passenger time consumption compared with the baseline.

4.4 Sample Operational Strategies

Two operational strategies are evaluated:

1. Time-based escalator direction control, which reallocates service capacity to match dominant passenger flow;

2. Crowd separation using physical barriers, which reduces flow conflicts in high-density corridors.

Baseline and optimized scenarios are compared using mathematical performance indicators.

4.5 The Execution of Experiment

To refine the proposed operational strategies, this study integrates high-fidelity 3D architectural modeling with multi-agent algorithmic simulations. In modeling, stations are reconstructed based on physical layout characteristics. These models are processed via AnyLogic 8.9, where crowd dynamics conservation principles are applied to simulate real-time passenger trajectories (Zhang et al., 2024). This "visual observation" methodology facilitates the identification of localized structural bottlenecks that purely analytical models may overlook. By iteratively testing interventions—such as time-based escalator direction control—within the simulated environment, the impact on total travel time t_i (mentioned in 4.3) and maximum platform density d_{i+1} (mentioned in 4.3) is visually verified and optimized.

Preliminary Suppositions

It is expected that time-based escalator direction control will reduce queue length and waiting time at bottlenecks by increasing effective service capacity in the dominant flow direction. Similarly, pedestrian flow separation is expected to lower local passenger density and improve corridor throughput.

Implications and Limitations

If validated, these results would suggest that significant congestion reduction can be achieved through operational optimization, without requiring major structural redesign of MRT stations.

This study, however, has several potential limitations. The most significant is that the mathematical model may not fully reflect real-world conditions, as certain factors may be overlooked or oversimplified. Another limitation is that the proposed optimization method is specifically designed for a single MRT station and may not be directly applicable on a larger scale. Nevertheless, the methodological framework underlying the optimization approach can be extended to other stations, thereby benefiting the overall MRT system.

Conclusion

This proposal outlines a mathematically grounded approach to MRT congestion mitigation that integrates empirical passenger-load data, established passenger-flow models, and

operational strategies. By focusing on Singapore's busiest MRT stations and evaluating interventions using quantitative indicators, the study aims to bridge the gap between theoretical congestion analysis and practical station operations.

References

Albi, G., Cristiani, E., Pareschi, L., & Peri, D. (2020). Mathematical models and methods for crowd dynamics control. In L. Gibelli (Ed.), Crowd dynamics: Theory, models and applications (pp. 159–197). Springer.
https://doi.org/10.1007/978-3-030-50450-2_8

Bahamid, A., & Mohd Ibrahim, A. (2022). A review on crowd analysis of evacuation and abnormality detection based on machine learning systems. Neural Computing and Applications, 34(24), 21641–21655.
https://doi.org/10.1007/s00521-022-07758-5

Hillier, F.S., & Lieberman, G. J. (2021). Introduction to Operations Research. McGraw-Hill Education.

Tan,YJ. (2025). Deep dive of Singapore's rail network passenger loads. Medium.
https://medium.com/@simpletan/deep-dive-of-singapores-rail-network-passenger-loads-e331d3b9626b

The Online Citizen. (2025, July 29). Commuters report heavy crowding at Circle Line MRT stations on 29 July morning.
https://www.theonlinecitizen.com/2025/07/29/commuters-report-heavy-crowding-at-circle-line-mrt-stations-on-29-july-morning/

World Transit Research. (n.d.). Research project 5465.
https://www.worldtransitresearch.info/research/5465/

Xu, X.-Y., Liu, J., Li, H.-Y., & Hu, J.-Q. (2014). Analysis of subway station capacity with the use of queueing theory. Transportation Research Part C: Emerging Technologies, 38, 28–43.
https://doi.org/10.1016/j.trc.2013.10.010

Zhang, J., Ai, Q., Ye, Y., & Deng, S. (2024). Dynamic flow analysis and crowd management for transfer stations: A case study of Suzhou Metro. Public Transport, 16(2), 619–653.
https://doi.org/10.1007/s12469-024-00357-8

Zitouni, M. S., Bhaskar, H., Dias, J., & Al-Mualla, M. (2016). Advances and trends in visual crowd analysis: A systematic survey and evaluation of crowd modelling techniques. Neurocomputing, 186, 139–159.
https://doi.org/10.1016/j.neucom.2015.12.070